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Adapted Koldo Belasko trap: a promising tool to reduce yellow-legged hornet pressure on beehives

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The European beekeeping sector is facing serious challenges due to the accidental arrival of the yellow-legged hornet (*Vespa velutina* Lepeletier, 1836). This invasive species undermines honeybee colonies and disrupts the workers' ability to forage effectively. Various control methods are currently used to manage this invasion. In this study, we evaluated the performance of a trap adapted from the design proposed by beekeeper Koldo Belasko, hereafter referred to as the Adapted Koldo Belasko (KBA) trap. We set up KBA traps across several apiaries within the Girona area (Catalonia, northeastern Spain) during 2023. The traps consistently captured high numbers of *V. velutina* and showed relatively high selectivity regarding non-target captures, particularly in the case of species smaller than *V. velutina*. These findings are highly relevant to the beekeeping sector and regulatory authorities, as they highlight how the KBA trap can help to prevent colony losses. Overall, the KBA trap is presented as a sustainable and environmentally friendly, for integrated hornet management in beekeeping.

Keywords: apiaries, control method, KBA trap, sustainable beekeeping, yellow-legged hornet

Introduction

Biological invasions greatly endanger biodiversity, as they can change the makeup of native communities and disrupt how natural ecosystems operate (Brondizio et al. 2019). Invasive alien species can have significant social and economic effects, as seen in European beekeeping with the arrival of the yellow-legged hornet (*Vespa velutina nigrithorax*, Buysson 1905) (Monceau et al. 2014, Laurino et al. 2019). Since its introduction to France in 2004, this species has rapidly expanded its range by several kilometers annually due to the high reproductive and dispersal capacity of founding queens and it is now established in several European countries (Arca et al. 2015, Rome et al. 2015). Although there are no specific studies on the actual dispersal abilities of future founding queens, Sauvard et al. (2018) measured hornet flight capacities, suggesting a potential for long-distance dispersal. Consequently, it has been listed as an invasive alien species of Union Concern by the European Union (European Commission 2024).

The yellow-legged hornet often preys on honeybees (*Apis mellifera*), its larvae's main food source (Villemant et al. 2011, Verdasca et al. 2022, Pedersen et al. 2025). Honey bee colonies face significant losses when hornets capture bees near hive entrances. This results in defensive blocking behavior, preventing bees from exiting the hive and decreasing their foraging activity (Arca et al. 2014, McClenaghan et al. 2019, Barbet-Massin et al. 2020). Prolonged periods of such behavior can severely weaken colonies, negatively impacting honey and pollen production, colony health, and increasing susceptibility to other stressors and diseases affecting bees (Potts et al. 2010, Requier et al. 2019b).

Although preventing the introduction of invasive species is the most effective strategy (Kenis et al. 2009), mitigation measures at the apiary level become essential once the species becomes established. Various strategies have been used throughout Europe to minimize their negative impact, including physically or chemically locating and destroying nests, as well as employing traps with differing levels of selectivity to capture queens and workers (Monceau et al. 2012, Rojas-Nossa et al. 2018, 2023, Lioy et al. 2020). Moreover, beekeepers also face difficulties in locating hornet nests when trees are in leaf (in the case of evergreen species or when deciduous trees retain their foliage), due to the extra time required to find these nests and the legal constraints associated with handling pesticides for their removal. Nevertheless, *in situ* measures within hives are the most effective and hold the greatest interest for beekeepers (Turchi and Derijard 2018, Pérez-Granados et al. 2023, Rojas-Nossa et al. 2023, Diéguez-Antón et al. 2024).

The most widely used physical methods include electric or manual rackets, metal mesh hive guards (muzzles), and electric

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harps (Pérez-Granados et al. 2023, Rojas-Nossa et al. 2023, Diéguez-Antón et al. 2024). More recently, innovations include individual electric mini-harps, electric harps combined with muzzles capable of capturing hornets, and tubular mesh traps such as the one developed by the beekeeper Koldo Belasko (known as the "Velutina control method", https://www.youtube.com/watch?v=QiK86ZzZWfU). These methods avoid the use of pesticides and align with ecological and sustainable beekeeping practices (Monceau et al. 2017).

In this study, we aim to evaluate a modified version of Koldo Belasko's trap (hereafter referred to as the adapted KB or adapted Koldo Belasko [KBA] trap) as a tool for yellow-legged hornet control in apiaries. Our specific objectives are to (i) examine the number of captures in the KBA trap and the phenology of the yellow-legged hornet and other hornets interacting with honeybee colonies; (ii) assess temporal changes in the size and caste composition of *V. velutina* in the KBA trap; and (iii) evaluate the selectivity of the KBA trap regarding non-target species. We also list the necessary materials for constructing a KBA trap, such as horizontal pipes, vertical chimneys, hoods, PVC profiles, caps, flanges, and hot-melt glue, but exclude tools from this list. This is the first scientific assessment of this method, which provides a practical, eco-friendly alternative to chemical treatments and maintains normal honeybee colony activity and productivity.

Materials and Methods

Adapted Koldo Belasko Trap

The KBA trap is an adaptation of a design originally developed by the beekeeper Koldo Belasco (presented at: https://www.youtube.com/watch?v=QiK86ZzZWfU). It consists of a horizontal meshed tube placed in front of the hive entrances, with vertical tubes at each end to facilitate the capture of yellow-legged hornets, while allowing honey bees to escape through the squared mesh openings (Fig. 1). To improve bee passage and reduce pollen loss—which was observed in a previous trial with the original 6 mm mesh—the horizontal tube was separated slightly from the hive entrances, and the mesh size was increased from 6 mm to 8 mm (further details are provided below). The modifications are designed to maximize hornet capture efficiency while minimizing interference with honeybee activity, drawing on empirical data from previous trials.

The horizontal tube has a 1-m perimeter and was constructed from galvanized wire mesh with 8 mm square openings, assembled by joining the marge with plastic flanges placed every 20 cm. At both ends of this tube, vertical tubes (referred to as chimneys) made of galvanized wire mesh with 6 mm square openings were attached. These chimneys are constructed from 1-m square sheets, shaped into cylinders with a 1-m perimeter and secured using plastic flanges at 10 cm intervals. Each chimney contains a trap collector at the top, consisting of a 4 mm thick plastic mesh funnel adhered to a PVC cylinder (12 cm in diameter, 10 cm in height) using thermal adhesive. A matching 4 mm thick plastic hood is fitted onto this assembly to contain the hornets once captured. Components such as the funnels and PVC tubes are pre-assembled in the laboratory, where the fit of the hoods is also verified. The remaining elements, horizontal tubes and chimneys, are assembled on site. Although the length of the horizontal mesh varies according to hive layout, it does not exceed 10 m to ensure structural integrity and effective trapping.

The horizontal tube is installed in front of the hive entrances, with an approximate separation of 8 cm, which is a key modification from the original design. This spacing preserves the natural flight path of honey bees and minimizes pollen loss during entry, since bees are not forced to pass through the 6 mm mesh, while also facilitating an easy exit from the hives. To allow hornet entry, rectangular windows (measuring 25 × 15 cm) are cut along the front, lower half of the horizontal tube at 1-m intervals. These openings enable yellow-legged hornet individuals to enter the trap as they attempt to prey on honey bees at the hive entrance. Once inside the horizontal tube, hornets typically move along the length of the structure toward the ends, ascending into the vertical chimneys. There, they are funneled into the plastic collection chambers, which effectively contain them. The rigidity of the plastic hoods also permits

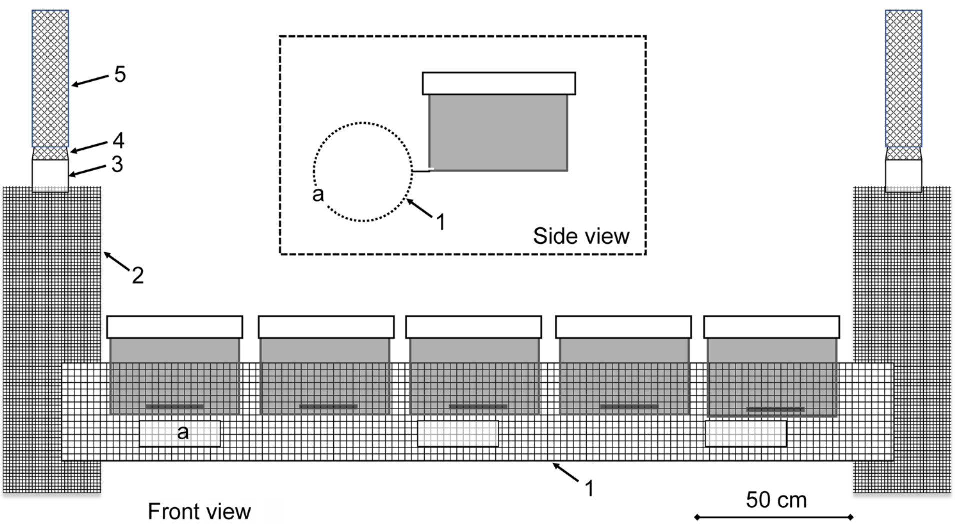

Fig. 1. Diagram of the front and side views of a hypothetical apiary with 5 Langstroth beehives protected with the KBA trap. The main components are: (a) horizontal tube with 8 mm mesh and windows (f); (b) vertical tube with 6 mm mesh; (c) PVC cylinder; (d) plastic funnel; and (e) plastic hood.

mechanical neutralization of live individuals during monitoring. Honey bee flight behavior at the hive entrance can occur in different ways (Fig. 1). They may enter and exit through the windows, pass through the 8 mm mesh, or fly over or under the horizontal tube.

Yellow-Legged Hornet Captures and Phenology in the KBA Trap

In 2023, we installed KBA traps in twelve apiaries across 3 different regions in the Girona area (Table 1) and monitored the number of hornets captures over time. All apiaries were located in areas where the presence of yellow-legged hornet had been previously confirmed, with varying levels of hornet pressure. The size of the apiaries ranged from 2 to 41 hives (Fig. 2), and *Varroa destructor* Anderson & Trueman (2000) infestation rates were assessed prior to the experiment to confirm that levels were low (<3%; measured in worker brood cells), due to the prior application of appropriate acaricidal treatments. This ensured that varroa was not a confounding factor, allowing the study to isolate the influence of *V. velutina* (Roura-Pascual et al. 2026). KBA traps were installed in most hives at the beginning of July, except for 2 transhumant apiaries (GAR1 and GAR6), where they were installed in August.

We recorded the number of captured yellow-legged hornets weekly, except in SEL1 and SEL2, where counts were conducted every 2 wk. To account for differences in sampling effort among apiaries, captures were standardized by the number of days that the traps remained active and the number of hives in each apiary, and were therefore expressed as captures per day per hive. Bar plots were used to represent both total number of captures and captures per hive in each apiary, while line graphs illustrated temporal trends over the sampling period across apiaries. In addition, we estimated the predation pressure by *V. velutina*, in apiaries in the absence of periodic removal via the KBA trap, assuming a conservative worker lifespan of 35 d based on Matsuura (1984) (Supplementary Text S1).

Caste Composition of Yellow-Legged Hornets in the KBA Trap

We also monitored the size of *V. velutina* individuals captured in the GAR5 apiary, protected with the KBA trap, in 2022 (Table 1). The specimens were preserved by freezing and subsequently analyzed morphometrically to determine caste composition. Specimens were identified by external morphology, as males can be distinguished from workers and queens by their lack of a sting and longer antennae (Diéguez-Antón et al. 2024). Following the methodology described by Perez de Heredia et al. (2017), mesoscutum width (MW) was measured to distinguish between workers and queens. Individuals measured with MW ≥4.5 mm were classified as queens, while those with MW <4.5 mm were categorized as workers. For morphometric analysis, pinned specimens were mounted on white polystyrene boards with a millimetric scale and photographed using a fixed JENOPTIK ProgResC10plus digital camera connected to a computer. MW measurements were then obtained using the ImageJ software platform (Rasband 1997–2016). Temporal changes in the size of captured hornets over the sampling period were examined using boxplots. In addition, a histogram of hornet sizes was generated, highlighting individuals with a maximum width (MW) greater than 0.45 cm, corresponding to the threshold used to differentiate queens from workers.

Non-Target Species Captured in the KBA Trap

In the 7 apiaries located in the Garrotxa region in 2023 (Table 1), data on non-target arthropods were also collected alongside *V. velutina* counts to assess the selectivity of the KBA trap and its potential impact on local biodiversity. These were classified into 8 categories: Honey bees (*Apis mellifera* Linnaeus 1758); European hornets (*V. crabro* Linnaeus 1758); African death's-head hawkmoths (*Acherontia atropos* Linnaeus 1758); other moths; bumblebees (*Bombus* spp.); day butterflies; orthoptera; and other arthropods. The European hornet (*V. crabro*), with its larger body size compared to the invasive *V. velutina*, was counted separately because it is a native species of conservation interest that plays an important ecological role as a generalist predator in some regions. Boxplots were created to compare non-target captures across apiaries, alongside bar plots depicting the proportion of captures in each apiary throughout the study period. Finally, we used a line plot to compare the phenology of *V. velutina* and *V. crabro* captures across the 7 apiaries. In this study, selectivity is understood as the relative efficiency of the KBA trap in capturing the target species compared to non-target species, rather than the complete absence of bycatch. This allowed a more detailed assessment of potential non-target impacts caused by the traps.

Table 1. Locations and characteristics of the apiaries monitored in 2022 and 2023

N	Code	Municipality	Hives	Region	
1	BE1	Castell-Platja d'Aro	2	Baix Empordà	2023
2	BE2	Castell-Platja d'Aro	14	Baix Empordà	2023
3	BE3	Sant Feliu de Guíxols	5	Baix Empordà	2023
4	SEL1	Sils	8	La Selva	2023
5	SEL2	Sta Coloma de Farners	20	La Selva	2023
6	GAR1	Montagut i Oix	40	Garrotxa	2023
7	GAR2	Olot	40	Garrotxa	2023
8	GAR3	Sant Joan les Fonts	40	Garrotxa	2023
9	GAR4	Vall de Bianya	24	Garrotxa	2023
10	GAR5	Santa Pau	41	Garrotxa	2022, 2023
11	GAR6	La Vall d'en Bas	18	Garrotxa	2023
12	GAR7	Santa Pau	32	Garrotxa	2023

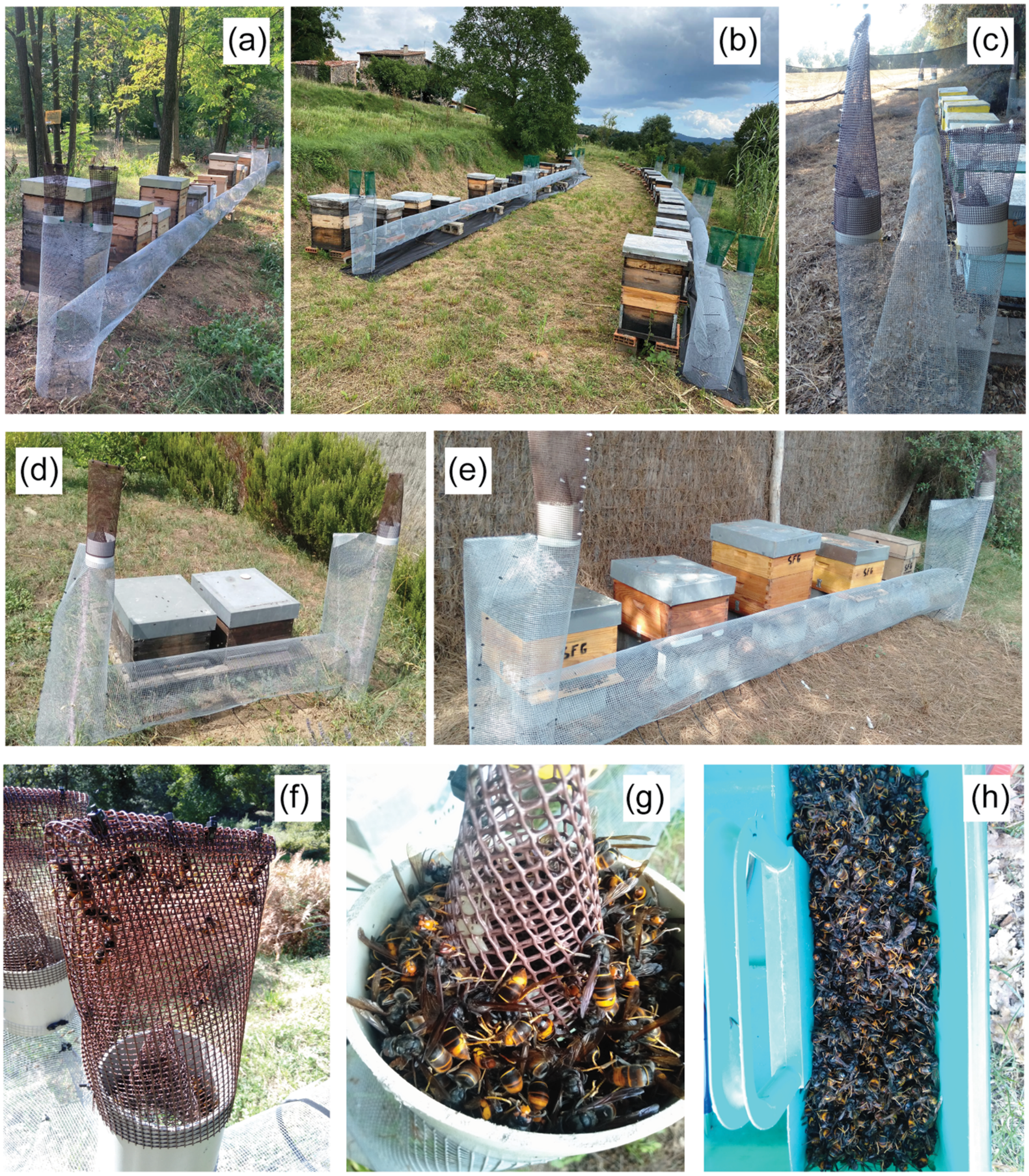

Fig. 2. Photographs showing KBA traps placed in front of hives at apiaries SEL2 (a), GAR2 (b), BE2 (c), BE1 (d), and BE3 (e); specimens of *V. velutina* captured in KBA traps, including live hornets captured inside the hood (f), dead hornets in the collector (g), and the total number of hornets collected at the end of the sampling period (h). See Table 1 for details of the apiaries.

Results

Yellow-Legged Hornet Captures and Phenology in the KBA Trap

In 2023, hornet pressure varied substantially between apiaries; even geographically close apiaries within the same region exhibited markedly different levels of pressure. For instance, apiaries GAR2 and GAR1 located 5.6 km apart in Garrotxa, and BE3 and BE2 separated by 2.7 km in Baix Empordà, showed distinct hornet pressures (Fig. 3). Smaller apiaries experienced proportionally higher hornet pressure relative to larger ones, with an extreme case observed at BE1 (Fig. 2), comprising only 2 colonies and recording an average pressure of nearly 11 hornets captured per hive per day (Fig. 3).

Temporal phenology of hornet activity displayed notable variability in 2023. In Garrotxa and Selva regions, hornet pressure followed a unimodal pattern with a later seasonal onset. Conversely, Baix Empordà exhibited an earlier onset and a bimodal distribution of hornet pressure (Fig. 4). Specifically, in apiaries BE1 and BE3 (Baix Empordà), a first peak of predation pressure occurred between late July and late August, peaking in early August, followed by a second peak from late September to late October, peaking in early October. In contrast, apiary BE2 lacked this second peak, showing instead a progressive decline in pressure from August onward (Fig. 4).

Apiaries in Garrotxa and Selva generally exhibited a strongly marked unimodal pressure pattern compared to Baix Empordà,

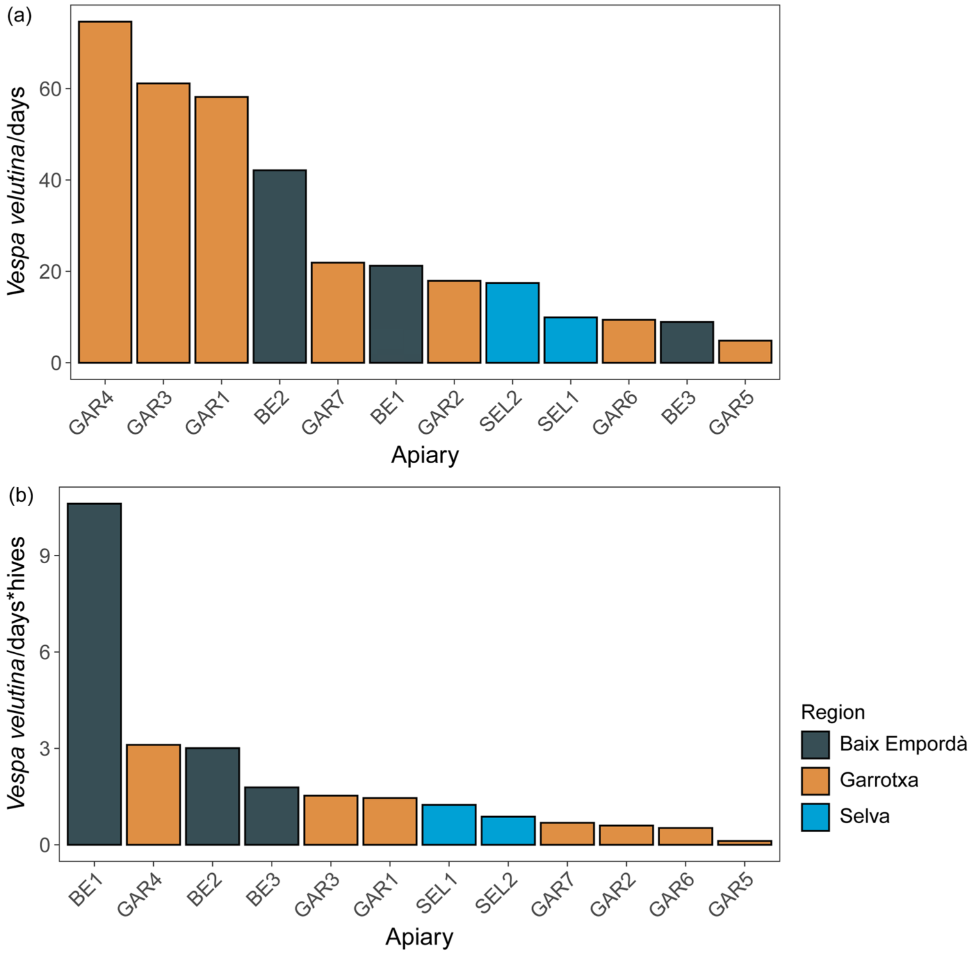

Fig. 3. Total number of yellow-legged hornets (*V. velutina*) captured in KBA traps across 12 apiaries (Table 1), expressed as captures per sampling days (a) and standardized by the number of hives and sampling days (b). Colors correspond to apiaries located in different regions.

with most apiaries presenting a medium to very high pressure (GAR2, GAR3, GAR5, GAR7, SEL1, SEL2) and only 3 apiaries presenting a low pressure (GAR1, GAR4, GAR6) (Fig. 3). In these regions, significant pressure began in mid-September and peaked in early October. These delayed phenological patterns relative to Baix Empordà likely reflect regional environmental conditions during summer. The unimodal pattern at GAR5 had been observed previously in 2022 with a similar timing (Supplementary Figs S1 and S3), though with a slight increase in captures in 2023 (8,130 hornets) compared to 2022 (7,554 hornets).

The total number of hornets captured daily tended to increase with apiary size (number of hives), but the pressure per individual hive decreased, albeit with a weak relationship (Supplementary Fig. S2). Consequently, smaller apiaries may experience disproportionately higher hornet pressure per hive (eg BE1), while larger apiaries show the opposite trend (eg GAR1) (Fig. 3). Without regular removal using the KBA trap, predation pressure from *V. velutina* was estimated to increase approximately 5 times, with cumulative hornet numbers per apiary ranging from 2,306 to 26,410 individuals depending on the apiary. Relative increases were greater in smaller apiaries, such as BE1, than in larger ones (Supplementary Fig. S3).

Caste Composition of Yellow-Legged Hornets in the KBA Trap

The KBA trap effectively deters yellow-legged hornets from approaching hive entrances, based on field observations, as well as facilitates the continuous removal of hornet individuals. In 2022, at the GAR5 apiary, 97.3% (7,554 individuals) of the trapped arthropods were identified as yellow-legged hornets. Among these, we detected an increase in MW over time, suggesting that the KBA trap could capture not only workers but also newly emerged founding queens and males later in the season (Fig. 5). Based on visual identification of males and an MW > 4.5 mm for queens (Perez de Heredia et al. 2017), the captured specimens consisted of 94.5% workers, 5.2% founding queens, and 0.3% males (Fig. 5). Notably, in November, putative founding queens accounted for up to 17% of total captures (Supplementary Fig. S4).

Non-Target Species Captured in the KBA Trap

Similar to the GAR5 apiary in 2022, where *V. velutina* represented 94.5% of total captures, this species was also by far the most captured across the 7 apiaries in the Garrotxa in 2023 (90.9 ± 8.1%), confirming consistent selectivity across different locations and seasons. Among the non-target species, those of similar or smaller size that can physically pass through the

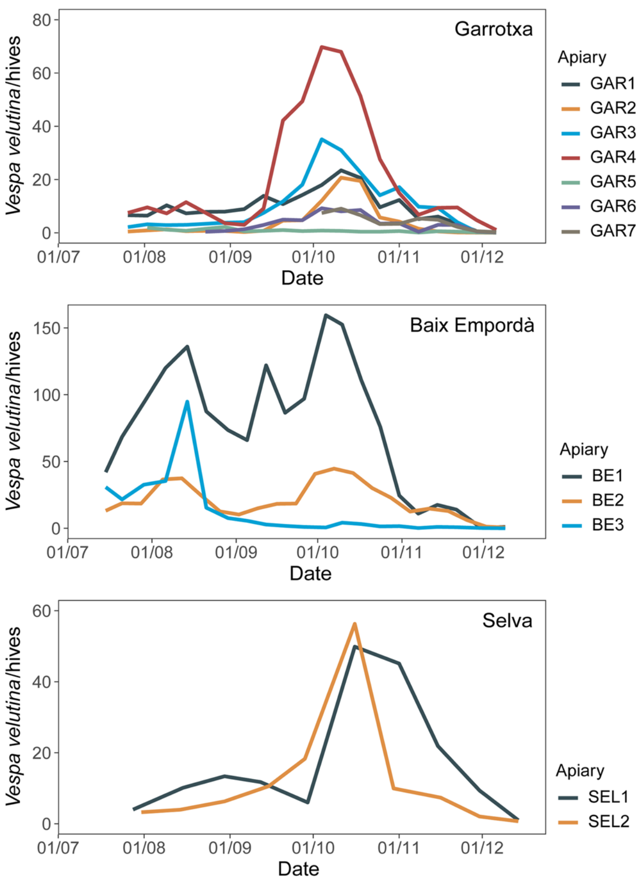

Fig. 4. Number of yellow-legged hornets (*V. velutina*) captured per hive in Adapted Koldo Belasko traps across 12 apiaries in 3 different regions: Garrotxa (top), Baix Empordà (middle), and Selva (bottom) (Table 1).

trap's funnel were occasionally captured. Contrarily, the European hornet (*V. crabro*) accounted for approximately 48 ± 25.9% of non-target captures (mean ± SD, $n = 7$). Other taxa, such as the African death's-head hawkmoth (*Acherontia atropos*) and various moths, were also trapped, despite being comparatively larger, likely due to their ability to enter the funnel with wings folded (Fig. 6).

Phenological monitoring of non-target species revealed variable temporal activity among non-target species. *Vespa crabro* exhibited a distinct phenology compared to *V. velutina*, with a peak of activity occurring between late September and early October (Fig. 7, Supplementary Fig. S5), similarly observed to GAR5 in 2022 (Supplementary Fig. S1). While there was a positive relationship in the overall number of captures between the 2 species in 2023, this relationship did not align with a linear trend line, indicating asynchronous seasonal dynamics (Supplementary Fig. S6).

Discussion

This study evaluates, for the first time, the KBA trap, a physical control method aimed at mitigating *V. velutina* pressure in managed honey bee apiaries. The modified KBA captures a high number of hornets and shows high selectivity with respect to non-target organisms, particularly for species smaller than *V. velutina* (Rojas-Nossa et al. 2023, Diéguez-Antón et al. 2024).

The modified KBA trap builds upon the original design proposed by beekeeper Koldo Belasko by optimizing mesh size and spatial configuration. The use of an 8 mm mesh, combined with positioning the trap approximately 8 cm in front of hive entrances, may improve colony access and reduces pollen loss, while its modular and lightweight design facilitates rapid deployment and straightforward maintenance under field conditions. Additional design features, such as peripheral capture points including vertical chimneys and collection hoods, appear to enhance passive hornet capture and contribute to high capture rates while minimizing disturbance to honey bee activity. By diverting hornets away from hive entrances toward these peripheral trap components, the KBA trap likely reduced direct hornet-bee interactions, which may benefit honey bee foraging behavior and colony vitality, although these effects were not directly quantified in the present study. Overall, the modified KBA trap not only yielded a high number of hornets captures but also strategically promoted spatial separation between hornets and hive entrances, thereby reducing colony disturbance and enabling uninterrupted foraging activity.

Comparison of multiple apiaries protected with KBA traps revealed marked spatial and temporal variability in hornet pressure (Monceau et al. 2014). Nearby apiaries experienced substantially different hornet pressure, although the underlying drivers of these differences remain unclear. Our results suggest that the size of apiaries may influence hornet pressure, with smaller ones experiencing a higher pressure per hive than larger ones. Environmental influences, although not directly tested, could help explain the phenological differences observed among monitored apiaries (Bessa et al. 2016, Rodríguez-Flores et al. 2019). For instance, regions such as Baix Empordà exhibited a bimodal pattern of hornet activity with early and late peaks, whereas others such as Garrotxa and Selva showed a unimodal distribution with later onset. These differences are likely influenced by climatic factors, including temperature and precipitation, as well as anthropogenic elements like urban nest removal. This regional heterogeneity highlights the need for localized adaptation of trap deployment timing to optimize control efficacy. Moreover, the potential impact of nest removal in urban environments, as documented near apiary BE3 with hornet numbers declining from mid-September onwards, underscores the importance of integrated management approaches combining trapping with nest detection and destruction.

The capture of worker hornets alongside putative founding queens by the KBA trap is a noteworthy observation. The 5-fold higher estimated pressure in apiaries in the absence of KBA protection highlights the intensity of hornet pressure in unprotected systems. The use of a conservative estimate of 35 d for hornet worker longevity, based on the general lifespan range of species in the genus *Vespa* (30 to 55 d; Matsuura 1984), provided a cautious yet ecologically grounded timeframe for interpreting hornet pressure during the sampling period. Additionally, the identification of queens based on the MW threshold following Pérez de Heredia et al. (2017), which has not been validated by examining hornet reproductive systems, should be interpreted with caution. If confirmed, the continuous removal of reproductive individuals, especially during the late-season period when queen production peaks, could contribute to population regulation of *V. velutina*. This dual function of providing immediate predation relief while

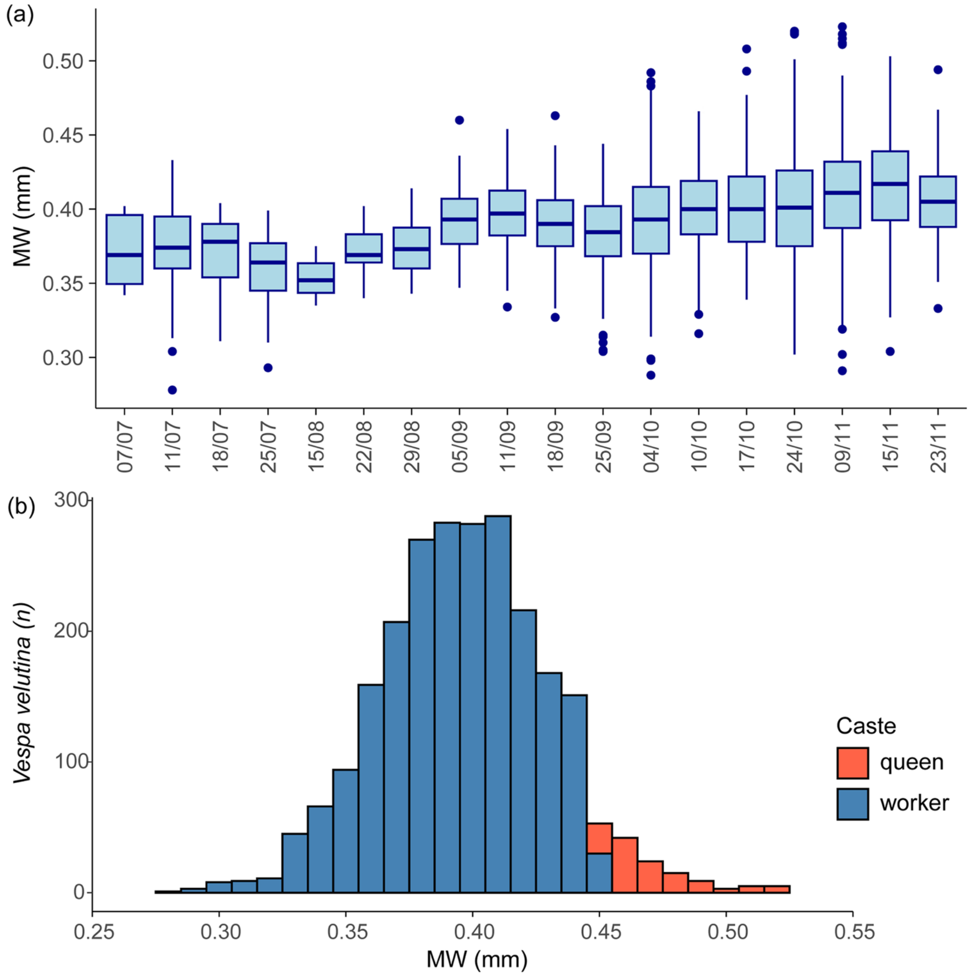

Fig. 5. Boxplots showing temporal variation in mesoscutum width (MW, in mm) of yellow-legged hornets (*V. velutina*, $n = 2{,}426$) captured at an apiary in Garrotxa region (GAR5; Table 1) in 2022 (a), together with a histogram of hornet size distribution (b). Boxplots in (a) represent the median and interquartile range (IQR), with whiskers extending to 1.5× IQR. Individuals with MW >4.5 mm, corresponding to the threshold used to distinguish queens from workers according to Heredia et al. (2017), are highlighted in red in panel (b).

simultaneously suppressing reproduction highlights the KBA trap's potential as a comprehensive management tool.

In addition to its capacity to capture hornets, the KBA trap also enables the collection of individuals suitable for post-capture morphometric analyses, which may provide valuable insights into caste structure and reproductive potential (Rome et al. 2015). The observed increase in worker body size observed over the season may reflect physiological and ecological changes linked to colony development and resource availability, potentially influencing foraging efficiency and predation capacity (Rome et al. 2015, De la Hera et al. 2023), although this interpretation remains tentative. Monitoring the demographic composition of captured hornets may therefore provide valuable insights into hornet population dynamics and inform optimized trap deployment strategies.

The KBA trap showed high selectivity toward *V. velutina*, with minimal bycatch of non-target species. Across sites and years, *V. velutina* accounted for approximately 95% or more of total captures. The observed bycatch can be partly explained by ecological and behavioral overlap, as several species exploit similar resources. Non-target captures mainly included species of similar or smaller size than *V. velutina* capable of passing through the trap's funnel. The European hornet (*V. crabro*) accounted for approximately 48% of non-target captures, while other taxa such as the African death's-head hawkmoth (*Acherontia atropos*) and various moth species were also occasionally trapped, likely due to their ability to enter with folded wings despite their larger size.

Phenological patterns, including distinct activity peaks and the absence of a relationship between *V. velutina* and *V. crabro*, are consistent with Monceau et al. (2015), who reported only partial overlap in the seasonal dynamics of the invasive and the native hornets. This temporal niche differentiation suggests that, despite sharing similar habitats and resources, the 2 species are not fully synchronous throughout the season. *Vespa crabro* plays an important role in native ecosystems and is considered less aggressive toward honey bees than *V. velutina* (Cini et al. 2018). Despite representing a relatively small proportion of total captures, the ecological significance of native species such as *V. crabro* underscores the need to further improve trap selectivity and minimize non-target impacts. Refinements to funnel aperture dimensions and trap deployment timing, aligned with the phenology of target and non-target species, may reduce bycatch and enhance conservation outcomes. Nevertheless, the selectivity observed for the KBA trap appears higher than that reported for other control

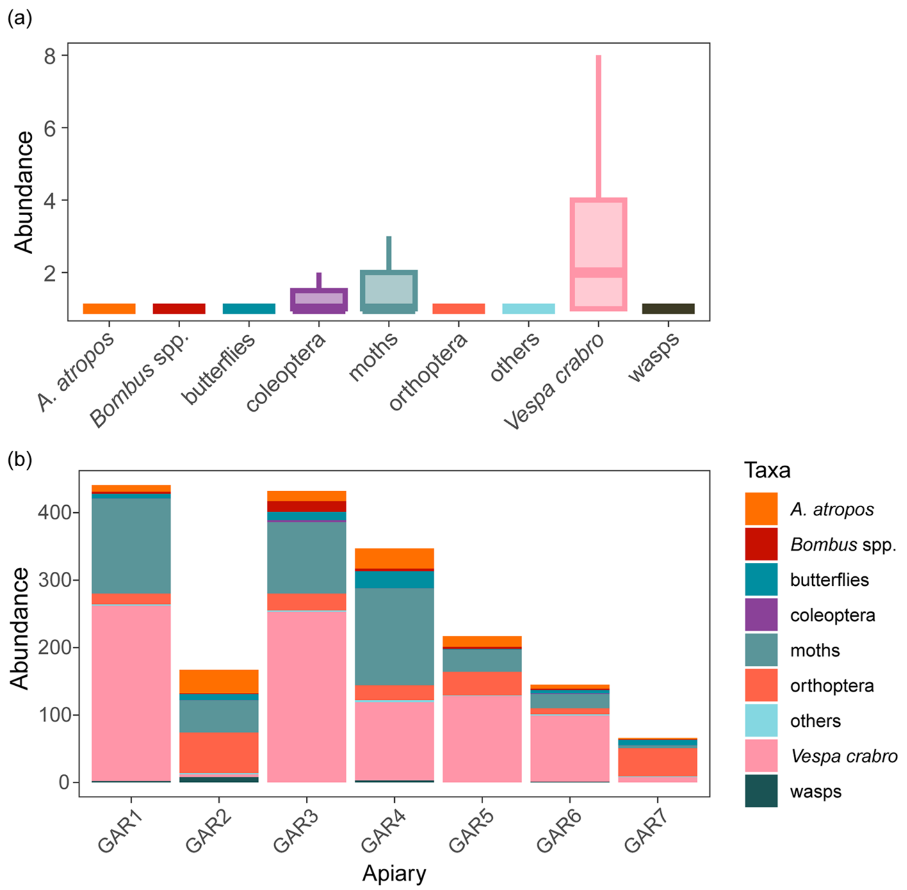

Fig. 6. Captures of non-target taxa in Adapted Koldo Belasko traps installed in 7 apiaries in Garrotxa region in 2023 (Table 1). (a) Boxplots showing daily captures of non-target taxa and (b) bar plots representing the proportion of non-target taxa captured in each apiary. Boxplots in (a) represent the median and interquartile range (IQR), with whiskers extending to 1.5× IQR. Abbreviation corresponds to *Acherontia atropos* (*A. atropos*).

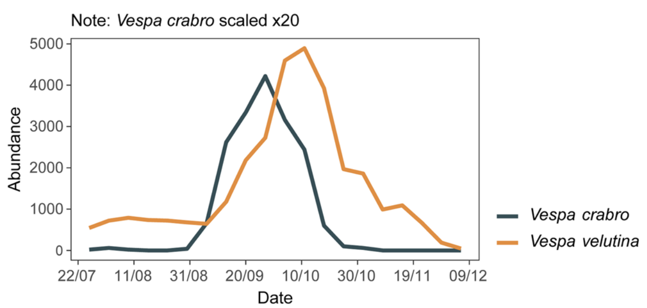

Fig. 7. Mean weekly captures of yellow-legged hornet (*Vespa velutina*, orange line) and European hornet (*V. crabro*, gray line) in Adapted Koldo Belasko traps at 7 apiaries in Garrotxa region in 2023. The values for European hornets have been scaled by a factor of 20 to allow comparison on a similar scale.

methods in previous studies (Lioy et al. 2020, Rojas-Nossa et al. 2023, Lueje et al. 2025).

Nevertheless, it should be acknowledged that there are certain limitations to this study that need to be considered. First, although field observations across the twelve apiaries protected with KBA trap suggested that honey bees remained active regardless of hornet pressure and that most colonies survived the winter period, these observations were not quantitatively assessed. Second, the use of mesoscutum width to distinguish gynes from workers, based on Pérez de Heredia et al. (2017), should be validated through examination of reproductive systems, as this has not yet been confirmed.

Besides these limitations, our results support the potential of mechanical trapping as a non-chemical, selective, and scalable alternative to traditional pest control methods. Preserving normal colony activity while effectively reducing hornet populations is critical for maintaining apiary productivity and ecological balance. The implementation of physical control methods like the KBA trap offers a promising path for sustaining honey bee colony health under invasive hornet pressure (Requier et al. 2019a). It is a sustainable alternative to chemical controls, which may have adverse effects on bees and the environment. The lack of detectable adverse effects on colony activity further supports the ecological compatibility of these methods. These findings suggest that hornet density, coupled with nest proximity, may constitute the primary factors influencing colony-level pressure, rather than apiary size alone.

Future research should prioritize refining trap designs to improve species-specific selectivity, and optimizing deployment strategies across diverse environmental conditions and considering the phenology of target and non-target species. Long-term studies incorporating standardized measures of hornet pressure and honey bee colony health, as well as further studies across multiple apiaries, and ecological contexts, are needed to validate and enhance integrated management protocols for *V. velutina*. Additionally, minimizing the impacts of KBA traps on non-target species populations is crucial to reduce unintended environmental effects, and the capture of reproductive gynes of *V. velutina* may also be a key factor in the management of this species. Overall, our findings suggest that the KBA trap is a promising and environmentally sustainable tool for reducing *V. velutina* predation pressure in managed honey bee apiaries.

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Supplementary Material

Supplementary material is available at *Journal of Economic Entomology* online.

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Conflicts of Interest

None declared.

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